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In [19]: import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import root_scalar
```

Exercise 3.2.1

Work out the least-squares solution when

$$A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \\ -2 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ -5 \\ 6 \end{bmatrix}$$

$$\begin{aligned} Ax - b &= \begin{bmatrix} 2 & -1 \\ 0 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 1 \\ -5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2x_1 - x_2 \\ x_2 \\ -2x_1 + 2x_2 \end{bmatrix} - \begin{bmatrix} 1 \\ -5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2x_1 - x_2 - 1 \\ x_2 + 5 \\ -2x_1 + 2x_2 - 6 \end{bmatrix} \\ A^T(Ax - b) &= \begin{bmatrix} 2 & 0 & -2 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2x_1 - x_2 - 1 \\ x_2 + 5 \\ -2x_1 + 2x_2 - 6 \end{bmatrix} = \begin{bmatrix} 8x_1 - 6x_2 + 12 \\ -6x_1 + 6x_2 - 6 \end{bmatrix} \stackrel{\text{SET}}{=} 0 \\ x &= \begin{bmatrix} -3 \\ -2 \end{bmatrix} \end{aligned}$$

Exercise 3.2.4

Prove that if A is an invertible square matrix, then $A^+ = A^{-1}$

$$A^+ = (A^T A)^{-1} A^T = (A^{-1} (A^T)^{-1}) A^T = A^{-1} ((A^T)^{-1} A^T) = A^{-1} I = A^{-1}$$

Exercise 3.3.4

Prove that if the $m \times n$, ($m \geq n$) matrix A is not rank-deficient, then the factor $\hat{R}$ of the thin QR factorization is invertible. (Hint: Suppose on the contrary that $\hat{R}$ is singular. Show using the factored form of A that this would imply that A is rank-deficient.)

Given some $A = \hat{Q}\hat{R}$ where $\hat{R}$ isn't invertible, WTS A isn't invertible.

Since $\hat{R}$ isn't invertible, $\det(\hat{R}) = 0$. Then $\det(A) = \det(\hat{Q}\hat{R}) = \det(\hat{Q}) \cdot 0 = 0$ means A isn't invertible.

Exercise 3.3.8

The matrix $P = \hat{Q}\hat{Q}^T$ derived from the thin QR factorization has some interesting and important properties.

(a) Prove that $P = AA^+$.

For sake of notation, let $\hat{Q} = Q$ and $\hat{R} = R$.

$$AA^+ = (QR)((QR)^T(QR))^{-1}(QR)^T = (QR)(R^{-1}Q^{-1})(Q^T)^{-1}(R^T)^{-1}(R^TQ^T) = Q(RR^{-1})(Q^TQ)^{-1}((R^T)^{-1}R^T)Q^T = QII^{-1}IQ^T$$

(b) Prove that $P^2 = P$. (This property defines a *projection matrix*.)

$$P^2 = (QQ^T)(QQ^T) = Q(Q^TQ)Q^T = QIQ^T = QQ^T = P$$

(c) Any vector x may be written trivially as $x = u + v$, where $u = Px$ and $v = (I - P)x$. Prove that u and v are orthogonal. (Together with part (b), this means that P is an *orthogonal projector*.)

First, know that $P^T = (QQ^T)^T = QQ^T = P$.

WTS $u^T v = 0$.

$$u^T v = (Px)^T(I - P)x = (x^T P^T - x^T P^T P)x = (x^T P - x^T P)x = 0$$

Thus u and v are orthogonal.

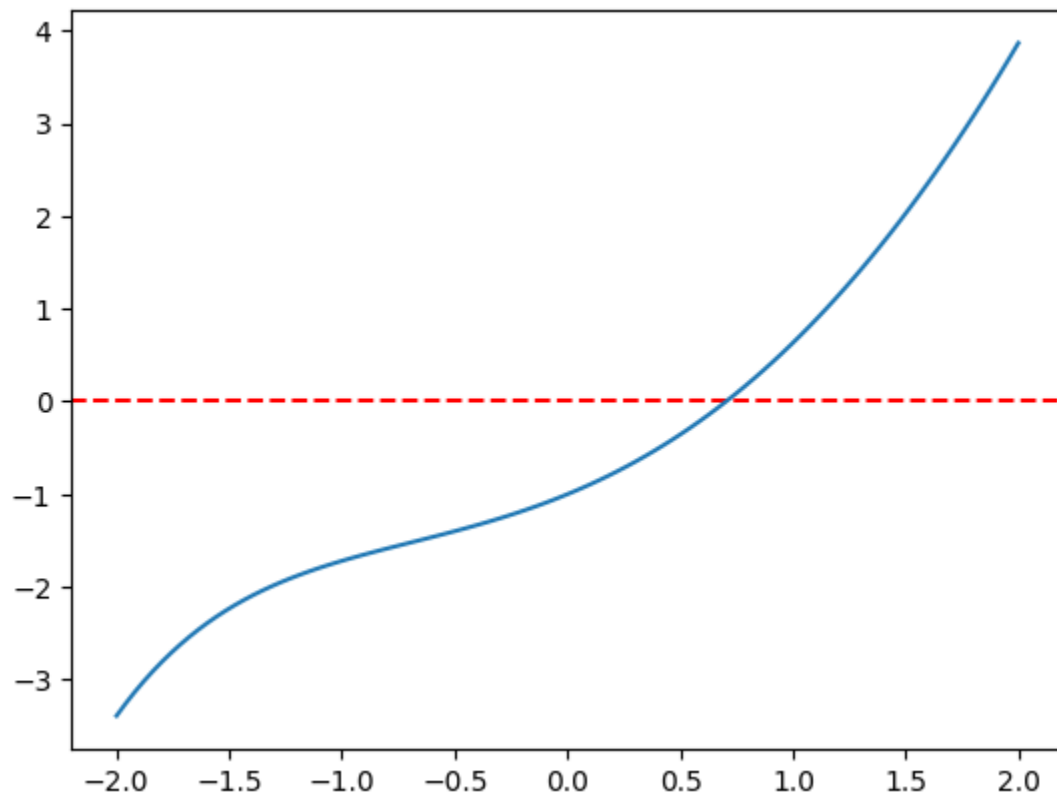
Exercise 4.1.1

$x^2 = e^{-x}$, over $[-2, 2]$

$$f(x) = x^2 - e^{-x} = 0$$

```
In [20]: def f(x):  
          return x ** 2 - np.exp(-x)  
  
x = np.linspace(-2, 2, 100)  
y = np.vectorize(f)(x)  
  
plt.axhline(0, linestyle = "--", color = "r")  
plt.plot(x, y)
```

Out[20]: [



Looks to be 1 root around $x = 0.7$.

```
In [21]: print(root_scalar(f, bracket = [0.5, 1]).root)
```

0.7034674224983917

$$\kappa_r = |2r + e^{-r}|^{-1}$$

$$\kappa_{0.703} \approx 0.526$$

Exercise 4.1.2

$$2x = \tan(x), \text{ over } [-0.2, 1.4]$$

$$f(x) = 2x - \tan(x) = 0$$

```
In [22]: def f(x):  
         return 2 * x - np.tan(x)
```

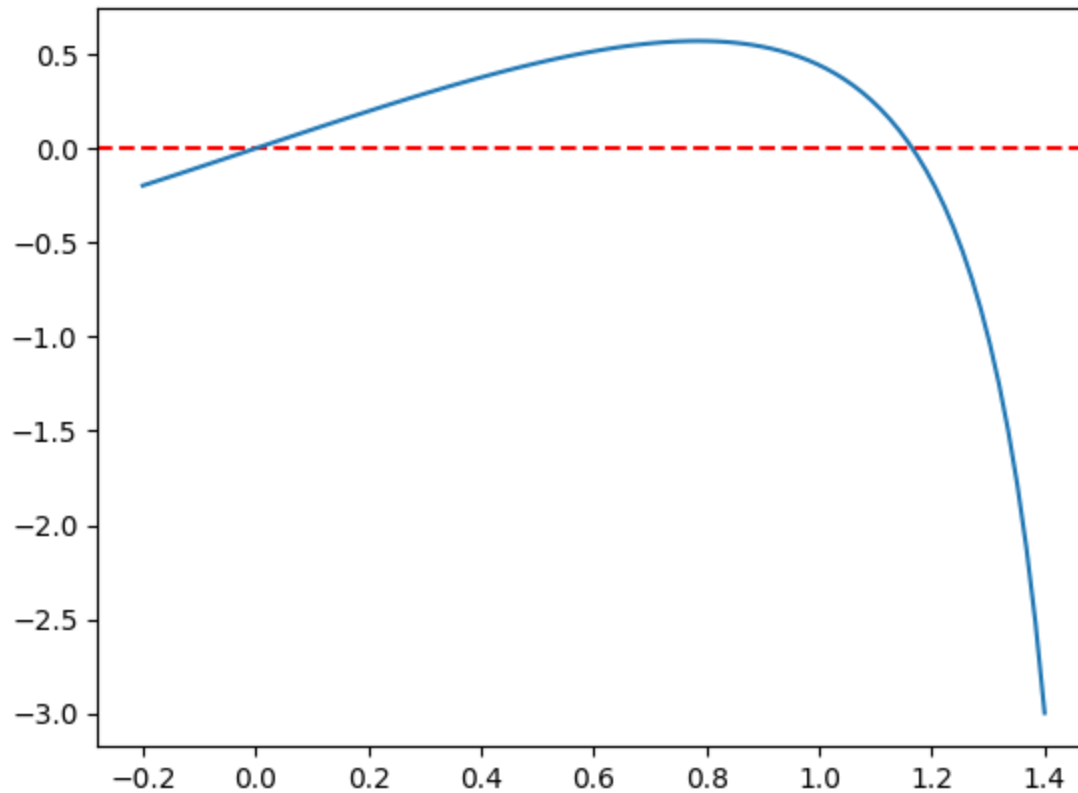
```

x = np.linspace(-0.2, 1.4, 100)
y = np.vectorize(f)(x)

plt.axhline(0, linestyle = "--", color = "r")
plt.plot(x, y)

```

Out[22]: [



Looks to be 2 roots around $x = 0, 1.2$.

```

In [23]: print(root_scalar(f, bracket = [-0.5, 0.5]).root)
          print(root_scalar(f, bracket = [0.8, 1.2]).root)

```

0.0

1.1655611852071506

$$\kappa_r = |2 - \sec^2(x)|^{-1}$$

$$\kappa_0 = 1, \kappa_{1.166} \approx 0.007$$

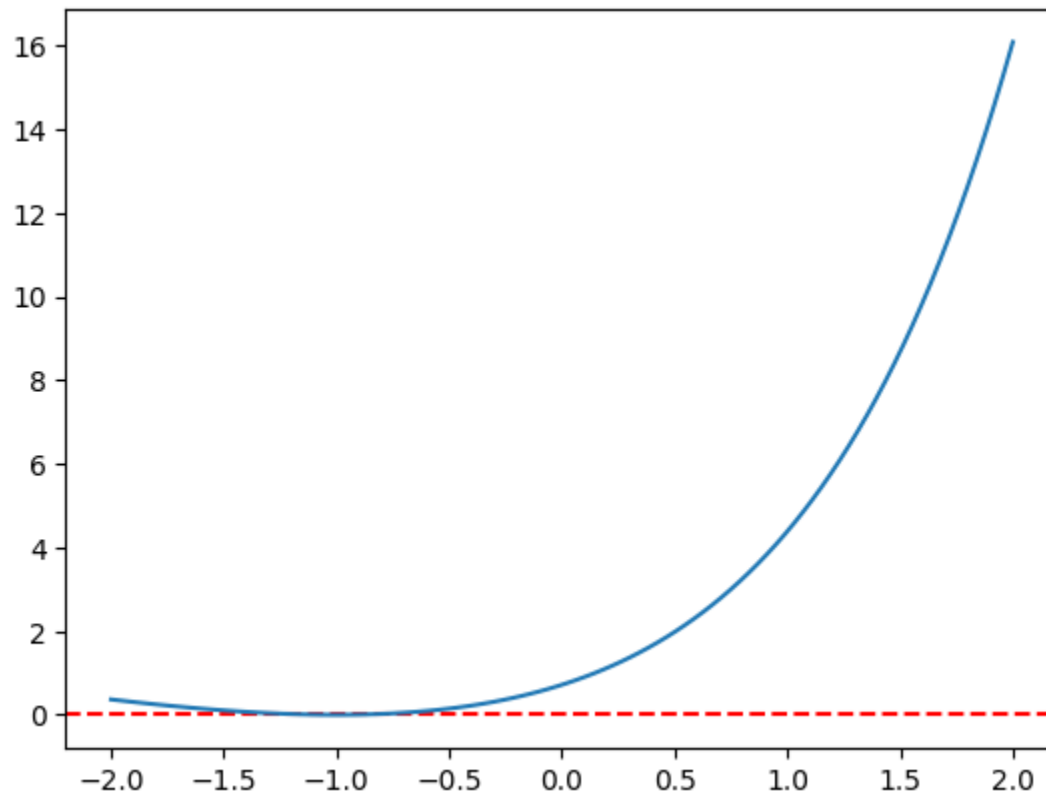
Exercise 1.4.3

$$e^{x+1} = 2 + x, \text{ over } [-2, 2]$$

$$f(x) = e^{x+1} - 2 - x = 0$$

```
In [24]: def f(x):  
         return np.exp(x + 1) - 2 - x  
  
x = np.linspace(-2, 2, 100)  
y = np.vectorize(f)(x)  
  
plt.axhline(0, linestyle = "--", color = "r")  
plt.plot(x, y)
```

Out[24]: [<matplotlib.lines.Line2D at 0x7f70b42ac590>]



I think there's only 1 root around $x = -1$, which is what Desmos says. We can't use the root finding method in example 4.1.1 because the tangent slope is 0 at $x = -1$

$$\kappa_r = |e^{x+1} - 1|^{-1}$$

κ_{-1} tends to infinity.

Exercise 4.1.7

For each function, find the multiplicity of the given root. If it is a simple root, find its absolute condition number.

(a) $f(x) = x^3 - 2x^2 + x - 2$, root $r = 2$

$f(x) = (x^2 + 1)(x - 2)$; multiplicity of 1. $|f'(2)|^{-1} = 1/5$

(b) $f(x) = (\cos(x) + 1)^2$, root $r = \pi$

f has a multiplicity of 4, evidenced by $f(x)/(x - \pi)^4 \neq 0, x = \pi$.

(c) $f(x) = \sin^2(x)/x$, root $r = 0$ (define $f(0) = 0$)

f has a multiplicity of 1, evidenced by $f(x)/(x - 0)^1 \neq 0, x = 0$. $\lim_{x \rightarrow 0} |f'(x)|^{-1} = 1$.

(d) $f(x) = (x - 1) \ln(x)$, root $r = 1$

f has a multiplicity of 2, evidenced by $f(x)/(x - 1)^2 \neq 0, x = 1$.